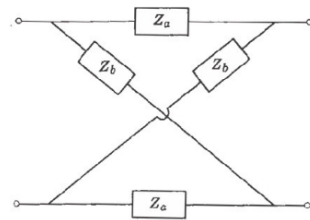
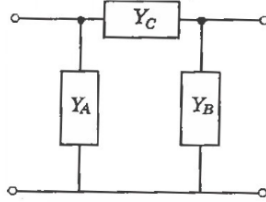
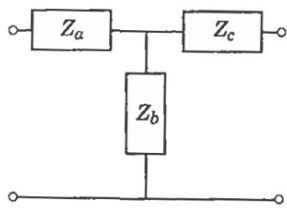
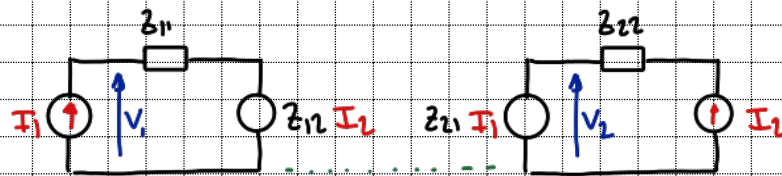


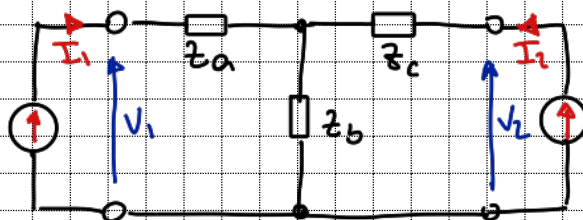
Tarea 6 - Montenegro, Maximiliano



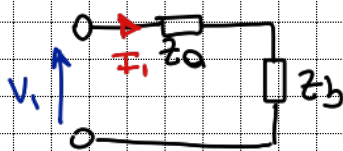
① $V = Z I$



$$\begin{cases} V_1 = Z_{11} I_1 + Z_{12} I_2 \\ V_2 = Z_{21} I_1 + Z_{22} I_2 \end{cases} \quad \begin{aligned} Z_{11} &= \left. \frac{V_1}{I_1} \right|_{I_2=0} & Z_{12} &= \left. \frac{V_1}{I_2} \right|_{I_1=0} \\ Z_{21} &= \left. \frac{V_2}{I_1} \right|_{I_2=0} & Z_{22} &= \left. \frac{V_2}{I_2} \right|_{I_1=0} \end{aligned}$$



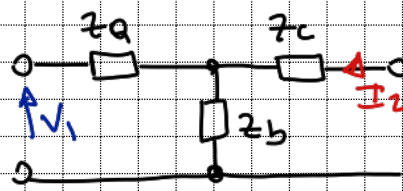
V_1 : con $I_2=0$



$$I_1 = \frac{V_1}{Z_a + Z_b}$$

$$\frac{V_1}{I_1} = Z_{11} = Z_a + Z_b$$

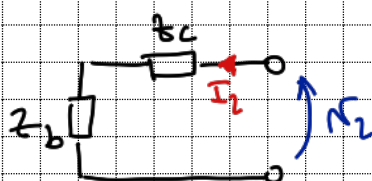
con $I_1=0$



$$I_2 = \frac{V_1}{Z_b}$$

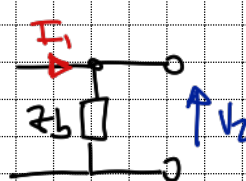
$$\frac{V_1}{I_2} = Z_{12} = Z_b$$

V_2 con $I_1=0$



$$\frac{V_2}{I_2} = Z_{22} = Z_b + Z_c$$

con $I_2=0$



$$\frac{V_2}{I_1} = Z_{21} = Z_b$$

$$z_b = z_{12} \quad z_a + z_b = z_{11} \Rightarrow z_a = z_{11} - z_{12}$$

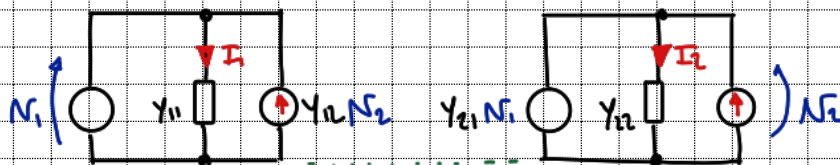
$$z_b + z_c = z_{22} \Rightarrow z_c = z_{22} - z_{12}$$

$$\begin{cases} V_1 = (z_a + z_b) I_1 + z_b I_2 \\ V_2 = z_b I_1 + (z_b + z_c) I_2 \end{cases}$$

$$\begin{cases} V_1 = (z_{11} - z_{12} + z_{12}) I_1 + z_{12} I_2 \\ V_2 = z_{12} I_1 + (z_{22} - z_{12} + z_{12}) I_2 \end{cases}$$

$$\begin{cases} V_1 = z_{11} I_1 + z_{12} I_2 \\ V_2 = z_{12} I_1 + z_{22} I_2 \end{cases}$$

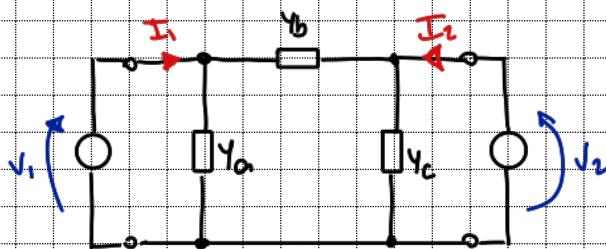
$$I = YV$$



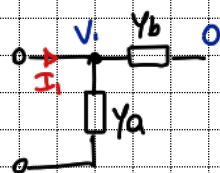
$$\begin{cases} I_1 = Y_{11} V_1 + Y_{12} V_2 \\ I_2 = Y_{21} V_1 + Y_{22} V_2 \end{cases}$$

$$Y_{11} = \left. \frac{I_1}{V_1} \right|_{V_2=0} \quad Y_{12} = \left. \frac{I_1}{V_2} \right|_{V_1=0}$$

$$Y_{21} = \left. \frac{I_2}{V_1} \right|_{V_2=0} \quad Y_{22} = \left. \frac{I_2}{V_2} \right|_{V_1=0}$$



$$I_1: \text{ con } V_2 = 0$$

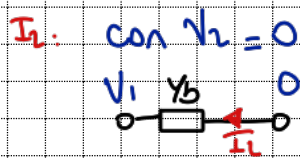


$$\frac{I_1}{V_1} = Y_{11} = Y_a + Y_b$$

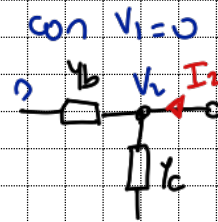
$$\text{ con } V_1 = 0$$



$$\frac{I_1}{V_2} = Y_{12} = -Y_b$$



$$\frac{V_1}{I_1} = Y_{11} = -Y_b$$



$$\frac{I_2}{V_2} = Y_{22} = Y_b + Y_c$$

$$Y_{11} = Y_a - Y_n$$

$$Y_a = Y_{11} + Y_n$$

$$Y_b = -Y_{21} = -Y_{12}$$

$$Y_{22} = -Y_{12} + Y_c$$

$$Y_c = Y_{12} + Y_{22}$$

$$\begin{cases} I_1 = (Y_{11} + Y_n - Y_n) V_1 + (-Y_n) V_2 \\ I_2 = (-Y_{21}) V_1 + (Y_n + Y_n - Y_n) V_2 \end{cases}$$

$$\begin{cases} I_1 = Y_{11} V_1 - Y_n V_2 \\ I_2 = -Y_n V_1 + Y_n V_2 \end{cases}$$

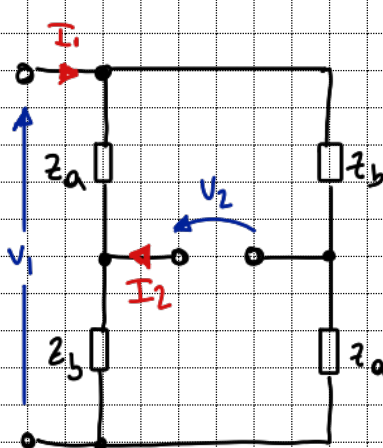
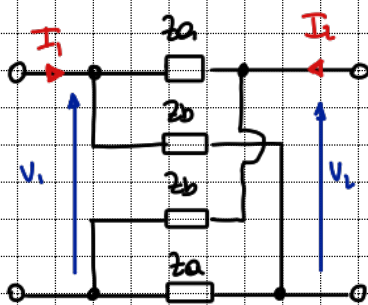
$$\begin{cases} V_1 = V_2 A + (-I_2) B \\ I_1 = V_2 C + (-I_2) D \end{cases}$$

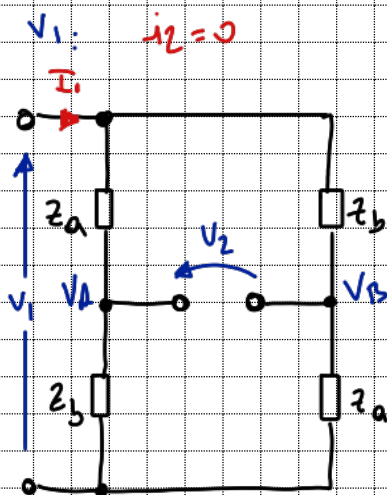
$$A = \left. \frac{V_1}{V_2} \right|_{I_2=0}$$

$$B = \left. \frac{V_1}{-I_2} \right|_{V_2=0}$$

$$C = \left. \frac{I_1}{V_2} \right|_{I_2=0}$$

$$D = \left. \frac{I_1}{-I_2} \right|_{V_2=0}$$





$$V_2 = V_A - V_B$$

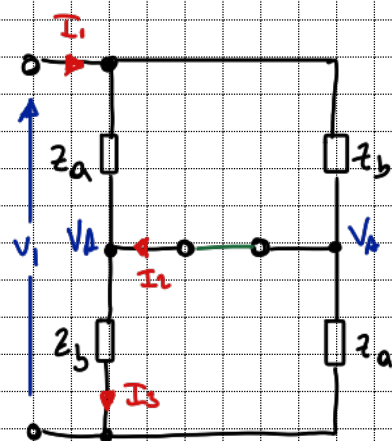
$$V_A = V_1 \frac{Z_b}{Z_a + Z_b}$$

$$V_B = V_1 \frac{Z_a}{Z_a + Z_b}$$

$$\Rightarrow V_2 = V_1 \frac{Z_b}{Z_a + Z_b} - V_1 \frac{Z_a}{Z_a + Z_b}$$

$$\frac{V_2}{V_1} = \frac{1}{A} = \frac{Z_b - Z_a}{Z_a + Z_b} \Rightarrow A = \frac{Z_a + Z_b}{Z_b - Z_a}$$

V_1 : $V_2 = 0$



$$I_3 = I_{11} + I_2 \quad I_{11} = \frac{V_1 - V_A}{Z_a} \quad V_A = \frac{V_1}{2}$$

$$I_3 = \frac{V_1}{Z_b}$$

$$I_2 = I_3 - I_{11}$$

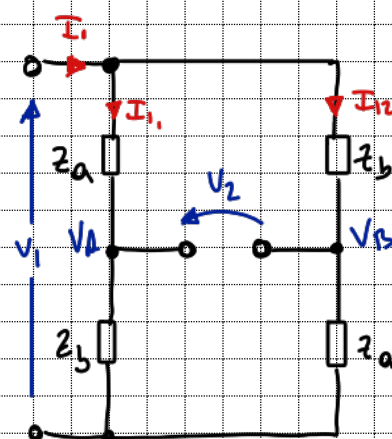
$$I_2 = \frac{V_1}{2Z_b} - \frac{V_1}{Z_a} + \frac{V_1}{2Z_a}$$

$$I_2 = V_1 \left(\frac{1}{2Z_b} - \frac{1}{2Z_a} \right)$$

$$I_2 = \frac{V_1}{2} \frac{Z_a - Z_b}{(Z_a Z_b)}$$

$$-\frac{V_1}{I_2} = B = \frac{2 Z_a Z_b}{Z_b - Z_a}$$

i_1 : $i_2 = 0$



$$I_1 = I_{11} + I_{12}$$

$$I_1 = \frac{V_1}{Z_a + Z_b}$$

$$I_{12} = \frac{V_1}{Z_a + Z_b}$$

$$\Rightarrow I_{11} = I_{12}$$

del parámetro B tengo

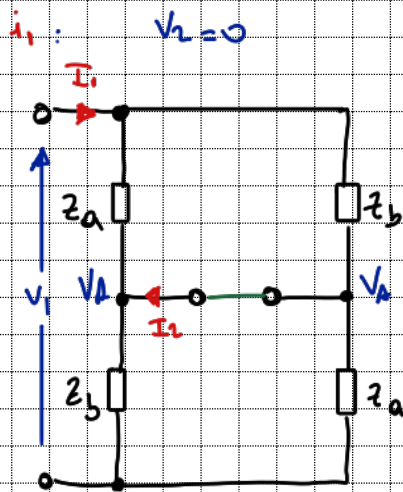
$$\frac{I_1}{2} = I_{11}$$

$$V_2 = V_1 \frac{Z_b}{Z_a + Z_b} - V_1 \frac{Z_a}{Z_a + Z_b}$$

$$V_2 = \frac{I_1}{2} Z_b - \frac{I_1}{2} Z_a$$

$$V_2 = \frac{I_1}{2} (Z_b - Z_a)$$

$$\frac{I_1}{V_2} = C = \frac{2}{Z_b - Z_a}$$



$$I_1 = I_{11} + I_{12} \quad I_{11} = \frac{V_1 - V_A}{z_a} \quad I_{12} = \frac{V_1 - V_A}{z_b}$$

$$I_1 = \frac{V_1 - V_A}{z_a} + \frac{V_1 - V_A}{z_b}$$

$$I_1 = V_1 \left(\frac{z_a + z_b}{z_a z_b} \right) - V_A \left(\frac{z_a + z_b}{z_a z_b} \right) \quad V_A = \frac{V_1}{2}$$

$$I_1 = V_1 \left(\frac{z_a + z_b}{z_a z_b} \right) - \frac{V_1}{2} \left(\frac{z_a + z_b}{z_a z_b} \right)$$

$$I_1 = \frac{V_1}{2} \left(\frac{z_a + z_b}{z_a z_b} \right)$$

Trayendo de B

$$I_2 = \frac{V_1}{2} \frac{z_a - z_b}{(z_a + z_b)}$$

$$V_1 = \frac{2 z_a z_b}{z_a - z_b} I_2$$

$$I_1 = \frac{2 z_a z_b}{z_a - z_b} \cdot \frac{1}{2} \left(\frac{z_a + z_b}{z_a z_b} \right) I_2$$

$$\frac{I_1}{-I_2} = D = \frac{z_a + z_b}{z_b - z_a} = A$$

$$Z = \begin{pmatrix} z_a + z_b & z_b \\ z_b & z_b + z_c \end{pmatrix} \quad Y = \begin{pmatrix} y_a + y_b & -y_b \\ -y_b & y_b + y_c \end{pmatrix} \quad T = \begin{pmatrix} \frac{z_a + z_b}{z_b - z_a} & \frac{2 z_a z_b}{z_b - z_a} \\ \frac{2}{z_b - z_a} & \frac{z_a + z_b}{z_b - z_a} \end{pmatrix}$$

$$\textcircled{2} \begin{cases} v_1 = A v_2 - I_2 B \\ I_1 = C v_2 - I_2 D \end{cases}$$

$$T \rightarrow z$$

$$C v_2 = I_1 + I_2 D \quad v_1 = \frac{A(I_1 + I_2 D) - I_2 B}{C}$$

$$v_2 = \frac{I_1 + I_2 D}{C} \quad v_1 = \frac{I_1}{C} + I_2 \left(\frac{AD - B}{C} \right)$$

$$v_2 = \frac{I_1}{C} + I_2 \frac{D}{C} \quad v_1 = \frac{I_1}{C} + \frac{I_2}{C} (AD - BC)$$

$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \frac{1}{C} \begin{pmatrix} A & AD - BC \\ 1 & D \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix}$$

$$V = I \cdot z$$

$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} \Rightarrow \begin{matrix} z_{11} = \frac{A}{C} & z_{12} = \frac{AD - BC}{C} \\ z_{21} = \frac{1}{C} & z_{22} = \frac{D}{C} \end{matrix}$$

$$T = \begin{pmatrix} \frac{z_a + z_b}{z_b - z_a} & \frac{2z_a z_b}{z_b - z_a} \\ \frac{2}{z_b - z_a} & \frac{z_a + z_b}{z_b - z_a} \end{pmatrix}$$

$$\frac{A}{C} = \frac{z_a + z_b}{z_b - z_a} \Rightarrow \frac{A}{C} = \frac{z_a + z_b}{2} = z_{11} \quad \frac{A}{C} = \frac{D}{C} = z_{22} = \frac{z_a + z_b}{2}$$

$$\frac{1}{C} = \frac{2}{z_b - z_a} = \frac{z_b - z_a}{2} = z_{21}$$

$$\begin{aligned} AD - BC &= \frac{(z_a + z_b)(z_a + z_b)}{(z_b - z_a)(z_b - z_a)} - \frac{2z_a z_b \cdot 2}{(z_b - z_a)^2} = \frac{z_a^2 + 2z_a z_b + z_b^2 - 4z_a z_b}{(z_b - z_a)^2} \\ &= \frac{z_a^2 - 2z_a z_b + z_b^2}{(z_b - z_a)^2} = \frac{(z_b - z_a)^2}{(z_b - z_a)^2} = 1 : AD - BC \end{aligned}$$

$$\Rightarrow \frac{AD - BC}{C} = \frac{1}{C} \Rightarrow z_{11} = \frac{z_b - z_a}{2}$$

$$\begin{cases} V_1 = z_{11} I_1 + z_{12} I_2 \\ V_2 = z_{21} I_1 + z_{22} I_2 \end{cases} \quad z \rightarrow Y \quad \begin{cases} I_1 = y_{11} V_1 + y_{12} V_2 \\ I_2 = y_{21} V_1 + y_{22} V_2 \end{cases}$$

$$\frac{V_1 - z_{11} I_1}{z_{12}} = I_2$$

$$V_2 = z_{21} I_1 + \frac{z_{22}}{z_{12}} (V_1 - z_{11} I_1)$$

$$V_2 - V_1 \frac{z_{22}}{z_{12}} = I_1 \left(z_{21} - \frac{z_{22} z_{11}}{z_{12}} \right) \Rightarrow z_{12} V_2 - z_{22} V_1 = I_1 (z_{12} z_{21} - z_{11} z_{22})$$

$$I_1 = \frac{1}{z_{12} z_{21} - z_{11} z_{22}} (z_{12} V_2 - z_{22} V_1), \quad z_{12} z_{21} - z_{11} z_{22} = -\Delta z$$

$$I_1 = -\frac{1}{\Delta z} (z_{12} V_2 - z_{22} V_1) \quad I_1 = \frac{1}{\Delta z} (V_1 z_{22} - V_2 z_{12})$$

$$I_2 = \frac{V_1}{z_{12}} - \frac{z_{11}}{z_{12}} \cdot \frac{1}{\Delta z} (V_1 z_{22} - V_2 z_{12})$$

$$I_2 = \frac{V_1}{z_{12}} - \frac{z_{11}}{z_{12}} \frac{z_{22}}{\Delta z} V_1 + \frac{z_{11}}{z_{12}} \frac{z_{12}}{\Delta z} V_2$$

$$I_2 = V_1 \left(\frac{\Delta z - z_{11} z_{22}}{\Delta z z_{12}} \right) + \frac{z_{11} z_{12}}{z_{12} \Delta z} V_2$$

$$I_2 = V_1 \left(\frac{z_{12} z_{22} - z_{12} z_{22} - z_{11} \Delta z}{\Delta z z_{12}} \right) V_1 + \frac{z_{11}}{\Delta z} V_2$$

$$I_2 = \frac{1}{\Delta z} (V_2 z_{11} - z_{21} V_1)$$

$$\begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \frac{1}{\Delta z} \begin{pmatrix} z_{22} & -z_{12} \\ -z_{21} & z_{11} \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix} \quad y_{11} = \frac{z_{22}}{\Delta z} \quad y_{12} = \frac{-z_{12}}{\Delta z}$$

$$y_{21} = \frac{-z_{21}}{\Delta z} \quad y_{22} = \frac{z_{11}}{\Delta z}$$

$$Y \longrightarrow T$$

$$\begin{cases} I_1 = Y_{11} V_1 + Y_{12} V_2 \\ I_2 = Y_{21} V_1 + Y_{22} V_2 \end{cases}$$

$\longrightarrow A$

$$\begin{cases} V_1 = A V_2 - I_2 B \\ I_1 = C V_2 - I_2 D \end{cases}$$

$$\frac{I_2 - Y_{21} V_2}{Y_{21}} = V_1$$

$$I_1 = Y_{11} \frac{1}{Y_{21}} I_2 - Y_{11} \frac{Y_{22}}{Y_{21}} V_2 + Y_{12} V_2$$

$$I_1 = \frac{Y_{12} Y_{21} - Y_{11} Y_{22}}{Y_{21}} V_2 + \frac{Y_{11}}{Y_{21}} I_2$$

$$Y_{11} Y_{21} - Y_{11} Y_{22} = -\Delta Y$$

$$\begin{cases} I_1 = -\frac{\Delta Y}{Y_{21}} V_2 + Y_{11} (-I_2) \\ V_1 = -\frac{Y_{12}}{Y_{21}} V_2 + \frac{1}{Y_{21}} (-I_2) \end{cases}$$

$$\begin{cases} I_1 = -\frac{\Delta Y}{Y_{21}} V_2 + Y_{11} (-I_2) \\ V_1 = -\frac{Y_{12}}{Y_{21}} V_2 + \frac{1}{Y_{21}} (-I_2) \end{cases}$$

$$\begin{pmatrix} I_1 \\ V_1 \end{pmatrix} = \frac{1}{Y_{21}} \begin{pmatrix} -\Delta Y & Y_{11} \\ -Y_{12} & 1 \end{pmatrix} \begin{pmatrix} V_2 \\ -I_2 \end{pmatrix}$$

$$A = -\frac{\Delta Y}{Y_{21}}$$

$$B = -\frac{Y_{11}}{Y_{21}}$$

$$C = -\frac{Y_{12}}{Y_{21}}$$

$$D = -\frac{1}{Y_{21}}$$